

A HASH-AND-SEPARABLE MATRIX OF ORDER 9

ABSTRACT. D'yachkov's lecture notes on designing screening experiments print a $C_{HS}(13, 4)$ -matrix and then ask, as Open problem 1, whether a $C_{HS}(q, 4)$ -matrix can be constructed with a smaller q . We exhibit a $C_{HS}(9, 4)$ -matrix, so that $q_4^{HS} \leq 9$. The matrix is printed in full; its three defining conditions reduce to nine 4×4 blocks and to 162 configurations on its 84 triples of rows. We claim no priority: whether such a matrix is already recorded in the literature is not determined here, and neither is whether $q_4^{HS} = 9$, which would need non-existence at $q = 7$ and $q = 8$.

1. THE QUESTION

Throughout, $q \geq k \geq 2$ are integers, $[q] = \{1, \dots, q\}$, and $*$ is a symbol not in $[q]$.

Definition. A $C(q, k)$ -matrix is a $q \times q$ matrix $C = (c_i(j))_{i,j \in [q]}$ with entries in $\{*\} \cup [q]$.

Such a matrix is the characteristic matrix of a code of length 3: its non- $*$ cells are read as the codewords $(c_i(j), i, j)$, and D'yachkov's Propositions 5, 6 and 7 [1] translate the three properties that code may have into conditions on the grid. We reproduce those three conditions in the form in which they are used below.

- (P5) Every symbol $a \in [q]$ occurs exactly k times in C ; every row and every column of C contains exactly $q - k$ stars; and no symbol occurs twice in one row or twice in one column.
- (P6) If $c_i(j) = c_m(n) = a \neq *$ with $i \neq m$ and $j \neq n$, then $c_i(n) = c_m(j) = *$.
- (P7) No 3×3 submatrix of C , on rows $i_1 < i_2 < i_3$ and columns $j_1 < j_2 < j_3$, has any of the six forms obtained by permuting the columns of

$$\begin{pmatrix} * & a & c \\ a & * & b \\ c & b & * \end{pmatrix}, \quad a, b, c \in [q] \text{ pairwise distinct.}$$

By Proposition 6, a $C(q, k)$ -matrix satisfying (P5) and (P6) is exactly a $C_H(q, k)$ -matrix (the hash case); by Proposition 7, one satisfying (P5), (P6) and (P7) is exactly a $C_{HS}(q, k)$ -matrix. The source supplies no grid condition for separable codes alone, and nothing below needs one: separability

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enters only through Definition 4 of the source, applied directly to a code, in the remark of Section 2. Following [1] we write q_k^{HS} for the least $q \geq k$ admitting a $C_{HS}(q, k)$ -matrix, that text asking verbatim

“How to find the minimal possible integer $q_k \geq k$ such that there exists $C_S(q_k, k)$ -matrix, $C_H(q_k, k)$ -matrix or $C_{HS}(q_k, k)$ -matrix?”

Example 10 of [1] then reads “Let $k = 4$. For hash codes, $q_4 = 8$ and for hash&separable codes, $q_4 = 13$ ”, and prints a $C_H(8, 4)$ and a $C_{HS}(13, 4)$ to support the two values. Immediately afterwards comes the display containing the question at issue:

“**Open problems.** 1. Is it possible to construct a $C_{HS}(q, 4)$ -matrix if $q < 13$? 2. Is it possible to construct $C_{HS}(q, k)$ -matrices, if $k \geq 5$ and $q < k^2$?”

Source. Quotations are from the file `lect_dse.tex` in the e-print source of `arXiv:1401.7505v1`; the printed lecture notes may number the propositions, examples and open problems differently, and nothing below depends on the labelling, since (P5)–(P7) and the question are reproduced in full above. The value 13 is the smallest order at which the source prints a $C_{HS}(q, 4)$ -matrix, and Open problem 1 asks, one paragraph later, for a smaller one; the source does not prove that none exists.

Related objects. Conditions (P5) and (P6) by themselves describe an object studied elsewhere under another name: a $q \times q$ array in which every row, every column and every symbol carries exactly k entries, and in which two equal entries in distinct rows and distinct columns force the two opposite cells to be empty, matches the definition of a *Blackburn k -plex* in [4], which proves a lower bound on the order of such an array and describes constructions for $k \leq 5$; (P7) is not among the conditions treated there. Füredi and Ruszinkó [3] characterise 3-union-free 3-partite linear hypergraphs by the avoidance of three grids, a counterpart of (P7); they give no value of q_4^{HS} , and they name small-value tables of Kim–Lebedev and of Lebedev for this family. The source states that the results of the section quoted above were to appear in [2]. We claim no priority: a $C_{HS}(q, 4)$ -matrix with $q < 13$ may already be recorded in that literature. What is offered here is the matrix and its verification.

2. THE MATRIX

Theorem 1. *The 9×9 matrix*

$$M = \begin{pmatrix} 1 & 2 & 3 & 4 & * & * & * & * & * \\ 5 & * & * & * & 9 & * & 4 & * & 8 \\ 6 & * & * & * & * & 8 & 3 & 2 & * \\ 7 & * & 9 & * & * & 4 & * & * & 2 \\ * & 8 & 6 & * & 1 & * & * & 4 & * \\ * & * & * & 9 & * & 1 & * & 7 & 3 \\ * & 9 & * & * & * & * & 1 & 5 & 6 \\ * & 7 & * & 6 & 3 & 5 & * & * & * \\ * & * & 5 & 8 & 2 & * & 7 & * & * \end{pmatrix}$$

is a $C_{HS}(9, 4)$ -matrix. Consequently $q_4^{HS} \leq 9$: a $C_{HS}(q, 4)$ -matrix with $q < 13$ exists, which is what Open problem 1 of [1] asks for.

For $a \in [q]$ let R_a and C_a be the sets of rows and of columns of M carrying a , and, for $i \in R_a$, let $\text{col}_a(i)$ be the column of a in row i . Reading M off:

a	R_a	C_a
1	1, 5, 6, 7	1, 5, 6, 7
2	1, 3, 4, 9	2, 5, 8, 9
3	1, 3, 6, 8	3, 5, 7, 9
4	1, 2, 4, 5	4, 6, 7, 8
5	2, 7, 8, 9	1, 3, 6, 8
6	3, 5, 7, 8	1, 3, 4, 9
7	4, 6, 8, 9	1, 2, 7, 8
8	2, 3, 5, 9	2, 4, 6, 9
9	2, 4, 6, 7	2, 3, 4, 5

(P5). Each row and each column of M carries exactly 4 symbols, hence exactly $9 - 4 = 5$ stars, and the four symbols in a row, and in a column, are distinct. Each $a \in [9]$ occupies the four cells listed in the table, so each occurs exactly 4 times; the $36 = 4 \cdot 9$ non-star cells are accounted for exactly once. $\square$

The next lemma is what makes (P6) a hand check: it replaces the quantifier over quadruples (i, j, m, n) by one $k \times k$ block per symbol.

Lemma 2. *Let C be a $C(q, k)$ -matrix satisfying (P5) and let $a \in [q]$. Then (P6) holds for the symbol a if and only if the $k \times k$ submatrix $C[R_a \times C_a]$ contains no non-star entry other than the k cells of a itself, i.e. if and only if that submatrix is a $k \times k$ permutation matrix over $\{a, *\}$.*

Proof. By (P5) the k cells of a lie in k distinct rows and k distinct columns, so $i \mapsto \text{col}_a(i)$ is a bijection $R_a \rightarrow C_a$; write row_a for its inverse. Let $i \in R_a$ and $n \in C_a$ with $n \neq \text{col}_a(i)$, and put $m = \text{row}_a(n)$. Then $m \neq i$, and $c_i(\text{col}_a(i)) = c_m(n) = a$ with $\text{col}_a(i) \neq n$, so (P6) forces $c_i(n) = *$.

Every cell of $C[R_a \times C_a]$ off the bijection is of this form, which gives one direction. Conversely, any instance of (P6) for the symbol a has $j = \text{col}_a(i)$ and $n = \text{col}_a(m)$ with $i, m \in R_a$ and $j, n \in C_a$, so the two cells $c_i(n)$ and $c_m(j)$ it constrains lie in $C[R_a \times C_a]$ off the bijection. $\square$

(P6). By Lemma 2 it suffices to inspect nine 4×4 blocks. For $a = 1$, $R_1 = C_1 = \{1, 5, 6, 7\}$ and $M[\{1, 5, 6, 7\} \times \{1, 5, 6, 7\}]$ has the four 1's on its diagonal and stars elsewhere. For $a = 2$, $R_2 = \{1, 3, 4, 9\}$, $C_2 = \{2, 5, 8, 9\}$, and the block carries 2 at $(1, 2), (3, 8), (4, 9), (9, 5)$ and stars in its other twelve cells. The remaining seven blocks are read off the same way from the table; each is a permutation matrix. $\square$

For (P7) we first remove the star conditions, which (P6) already implies. Write $\text{sh}(i, m)$ for the set of symbols occurring in both row i and row m .

Lemma 3. *Let C be a $C(q, k)$ -matrix satisfying (P5) and (P6). Then C fails (P7) if and only if there are rows $i_1 < i_2 < i_3$ and pairwise distinct symbols*

$$a \in \text{sh}(i_1, i_2), \quad b \in \text{sh}(i_2, i_3), \quad c \in \text{sh}(i_1, i_3)$$

with

$$(1) \quad \text{col}_a(i_1) = \text{col}_b(i_3), \quad \text{col}_a(i_2) = \text{col}_c(i_3), \quad \text{col}_b(i_2) = \text{col}_c(i_1),$$

these three columns being distinct.

Proof. Since the six forms of (P7) are the column permutations of one, and that one is symmetric, an occurrence on a set of three rows and a set of three columns may be taken with the rows in increasing order and the columns free. Such an occurrence, with columns j_1, j_2, j_3 , has a at (i_1, j_2) and (i_2, j_1) , b at (i_2, j_3) and (i_3, j_2) , and c at (i_1, j_3) and (i_3, j_1) ; reading off the columns gives (1) with $(j_1, j_2, j_3) = (\text{col}_a(i_2), \text{col}_a(i_1), \text{col}_b(i_2))$, distinct. Conversely, suppose (1) holds with j_1, j_2, j_3 as just named. The six named cells are then filled as required, and the three diagonal cells are stars: a sits at (i_1, j_2) and (i_2, j_1) with $i_1 \neq i_2$ and $j_1 \neq j_2$, so (P6) gives $c_{i_1}(j_1) = c_{i_2}(j_2) = *$; applying (P6) to b gives $c_{i_2}(j_2) = c_{i_3}(j_3) = *$ and to c gives $c_{i_1}(j_1) = c_{i_3}(j_3) = *$. So the submatrix on those rows and columns is a forbidden form. $\square$

(P7). M has $\binom{9}{2} = 36$ pairs of rows; exactly 18 of them share two symbols and 18 share exactly one, a total of $2 \cdot 18 + 1 \cdot 18 = 54 = q \binom{k}{2}$ shared incidences. In particular every one of the $\binom{9}{3} = 84$ triples of rows is pairwise sharing, so Lemma 3 must be applied to all of them. Running over the 84 triples and over the pairwise distinct choices of $(a, b, c) \in \text{sh}(i_1, i_2) \times \text{sh}(i_2, i_3) \times \text{sh}(i_1, i_3)$ leaves exactly 162 candidate configurations, and each of the 162 violates one of the three equations (1). Hence (P7) holds. $\square$

The 162 configurations are enumerated by the program described in Section 4, which also verifies (P7) on M with no reduction at all and re-derives every count quoted above. This completes the proof of Theorem 1.

Remark (the definitions applied directly). Read M 's 36 non-star cells as the codewords $(c_i(j), i, j)$. The resulting code of length 3 satisfies Definitions 2, 4 and 5 of [1] directly: each of the nine symbols occurs exactly 4 times in each of the three coordinates, any two codewords agree in at most one coordinate, every one of the $\binom{36}{3} = 7140$ triples has a coordinate on which all three differ, and the coordinatewise unions of the 7806 subsets of size at most 3 are pairwise distinct. So M is a $C_{HS}(9, 4)$ -matrix by the raw definitions and not only by way of Propositions 5–7.

3. WHAT IS AND IS NOT SETTLED

Lemma 4. *Every $C_H(q, k)$ -matrix, hence every $C_{HS}(q, k)$ -matrix, has $q \geq 2k - 1$.*

Proof. Fix $a \in [q]$. By Lemma 2 each of the k rows in R_a already carries $k - 1$ stars inside $C[R_a \times C_a]$, while by (P5) a row carries exactly $q - k$ stars in total. So $k - 1 \leq q - k$. $\square$

For $k = 4$ Lemma 4 gives $q \geq 7$, so $q = 7$ and $q = 8$ are the orders below 9 that this note does not decide.

What is settled is $q_4^{HS} \leq 9$, by finite checks on the printed matrix. The exact value is not: $q_4^{HS} = 9$ would additionally require that no $C_{HS}(7, 4)$ - and no $C_{HS}(8, 4)$ -matrix exists, and neither non-existence is attempted here or by the accompanying program. Open problem 2 of [1] ($k \geq 5$, $q < k^2$) is untouched.

4. REPRODUCTION AND VERIFICATION

Nothing in Theorem 1 needs a computer to be *stated*: the matrix is printed above, (P5) and (P6) are read off it by way of the table and Lemma 2, and (P7) reduces by Lemma 3 to 162 configurations. The program `verify.py` in this folder re-derives it all from the printed object, run as `python3 verify.py` with the standard library only and in exact integer arithmetic: (P5) clause by clause; (P6) both by exhaustive search over the 2304 quadruples (i, j, m, n) and by the nine blocks of Lemma 2; (P7) both by the reduction of Lemma 3 (162 candidates, none surviving) and by unreduced search over row and column triples; and the definitional route of the remark above, including all 7140 triples and all 7806 subsets. Each check is also run on the labelled matrices of Examples 7–10 of [1], which the source itself marks as satisfying some of the three conditions and failing others. The treatments of (P7) are encodings inside one program and share one transcription of M , so their agreement bears on the encodings and is not an independent confirmation. The recorded run `verify.output.txt` reports 50 checks, all passing, and lists in its own output what it does not cover — in particular the non-existence claims at $q = 7$ and $q = 8$, which it does not attempt.

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